

Uniform Mixing in Barrington Branching Programs

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Abstract. We study the distribution D_T over the symmetric group S_5 induced by evaluating any fixed valid balanced transposition-template Barrington branching program for the point function CC_T on uniformly random inconsistent inputs. We prove two results for this template. First, $D_T = D_{T'}$ for all $T, T' \in \{0, 1\}^\ell$ (exact T-independence), via a combinatorial flip bijection that requires no properties of the underlying group. Second, $d_{TV}(D_T, \text{Uniform}(S_5)) \leq \text{negl}(\ell)$ (spectral contraction), via Fourier analysis on S_5 using the Diaconis–Shahshahani Upper Bound Lemma. The contraction proof proceeds by exact computation of Fourier norms for the standard representation $(4, 1)$ at the first two levels of the commutator tree, followed by exhaustive verification that the operator norm drops below 1 at the third level — enabled by the discovery that the level-2 obstruction has a regular simplex structure in $\mathbb{R}^4$ that the commutator necessarily breaks. We further identify obstructions to extending this uniform contraction beyond S_5 : for S_n with $n \geq 6$ a dimensional obstruction forces every level-2 operator to have norm 1. At level 3, the maximum norm is 1 for S_6 and S_7 , although some configurations contract; for $n \geq 10$ every configuration has norm 1 by a dimensional bound. The analysis presented here does not decide the intermediate cases $n \in \{8, 9\}$. The two results combine into a quantitative statement about one exactly specified experiment: the q -sample transcript of terminal outputs obtained from fresh, independent, uniformly random inconsistent paths is exactly target-independent and within $\min\{1, q\varepsilon(\ell)\}$ of uniform, where $\varepsilon(\ell)$ is our explicit single-sample bound (so the transcript remains negligibly close to uniform whenever $q = q(\ell)$ is polynomially bounded), and both properties survive target-independent post-processing. This gives an unconditional, quantitative terminal-output guarantee for the fixed transposition-template Barrington layer.

Keywords: Barrington branching programs · spectral contraction · Fourier analysis on finite groups · symmetric group · indistinguishability obfuscation · point functions

1 Introduction

Indistinguishability obfuscation (iO) for general circuits is one of the most powerful primitives in cryptography, implying a vast array of applications from functional encryption to deniable encryption [GGH⁺13, SW14]. The first candidate constructions [GGH⁺13] were based on matrix branching programs over multilinear maps, randomized via Kilian’s technique [Kil88], following the classical correspondence established by Barrington’s theorem [Bar89]: every NC^1 circuit of depth d can be simulated by a branching program of width 5 and length at most 4^d over the symmetric group S_5 .

Obfuscation of branching programs, and the mixed-input attacks studied against it [CVW18], motivate the question we ask here: what distribution does such a program

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produce when it is evaluated on “incorrect” inputs? Specifically, for a point function $\text{CC}_T : \{0,1\}^\ell \rightarrow \{0,1\}$ that outputs 1 only on $x = T$, we study the fixed balanced transposition-template program BP_T described below: it produces a designated permutation $\pi_{\text{accept}} \in S_5$ on input T and the identity e on all other inputs. When this program is evaluated on a random bit string $b \in \{0,1\}^L$ that is *inconsistent* (at least one variable receives different values at different steps), the output is a random element of S_5 whose distribution we denote D_T .

Two properties of D_T are natural to ask of this layer, and are the ones we establish:

- (i) **T-independence:** $D_T = D_{T'}$ for all T, T' : the marginal law of the terminal output on a uniformly random inconsistent path is *exactly* independent of the target.
- (ii) **Uniform mixing:** $d_{\text{TV}}(D_T, \text{Uniform}(S_5)) \leq \text{negl}(\ell)$, so that marginal is statistically close to uniform on S_5 .

The experiment we study. Mixed-input evaluation—running a branching program on inputs that read a variable inconsistently—is the setting in which the algebraic behaviour of the S_5 layer is examined [CVW18], and it is what motivates the question we ask. We fix one exactly specified experiment and analyse it in full: the q -sample *terminal-output transcript*, in which each sample is the final permutation produced by BP_T on a fresh, independent, uniformly random inconsistent path (Definition 7). For the fixed balanced transposition template we prove that this transcript is *exactly* target-independent and within $\min\{1, q\varepsilon(\ell)\}$ of uniform on S_5^q , where $\varepsilon(\ell)$ is the explicit single-sample bound of Theorem 10(b); in particular the transcript stays negligibly close to uniform whenever $q = q(\ell)$ is polynomially bounded. Both properties are preserved by any target-independent post-processing of the transcript (Corollary 1).

Uniform sampling conditioned on inconsistency is an analytical average-case choice: it isolates the distributional behaviour of the fixed transposition-template Barrington layer outside consistent Boolean evaluations. We are not aware of an existing obfuscation security analysis that induces exactly this sampling law. The contribution is therefore a mathematical result about Barrington programs motivated by obfuscation.

A construction-level illustration. The GGH-style candidate of Garg, Gentry, Halevi, Raykova, Sahai, and Waters [GGH⁺13, full version, Sections 4.1–4.2] illustrates the objects a security analysis must relate. Its public garbled program contains encodings of randomized step matrices for the available input choices, encoded endpoint vectors, a parallel dummy program, and public encoding parameters. For an unfixed input bit both choices are available; evaluation combines these encodings and tests the resulting expression. An adversary receives this entire collection and can form expressions across evaluations.

Our theorem controls the terminal permutation of the plaintext template defined here under fresh uniform inconsistent-path sampling. This supplies a model for one underlying group-product distribution, without identifying our template with the branching program of that candidate. A cryptographic application would require a candidate-specific reduction relating the encoded view and its permitted operations to this sampling experiment, and controlling information beyond the terminal product. The terminal marginal alone does not establish statistical obfuscation or the security of an iO construction.

Our contributions. We prove both properties for every fixed valid transposition template (Definition 2).

For T-independence, we give a short combinatorial proof (Theorem 3) via a flip bijection F_S that maps inconsistent paths for T to inconsistent paths for T' while preserving the output permutation. The proof uses no property of S_5 and extends to conjunctions with wildcards.

For uniform mixing, we develop a Fourier-analytic framework on S_5 (Section 4). Using the Diaconis–Shahshahani Upper Bound Lemma, we reduce the problem to bounding the Fourier norms of the unconditional distribution P_T in each irreducible representation, then transfer the bound to D_T via a conditioning lemma (Lemma 1). The standard representation $(4, 1)$ of dimension 4 is the bottleneck; we therefore focus the exposition on it, while Proposition 1 and the level-3 certificate control all remaining non-trivial irreducibles, as the Diaconis–Shahshahani bound requires. We establish contraction at the first two levels of the commutator tree by exact symbolic computation (Theorems 4 and 5), discover that the level-2 obstruction has a regular simplex structure (Observation 6), and prove that this structure is necessarily broken at level 3 (Theorem 8). An exact integer Gram-form certificate over the full ordered universe then yields operator-norm contraction with $\lambda \leq 0.0355$. An inductive argument then gives doubly-exponential decay in the depth (Theorem 9).

The mixing bound holds for powers of two $\ell \geq 8$; the sampling hypotheses are collected in Remark 7. T-independence also extends to conjunctions with a fixed wildcard set. Section 6 presents evidence for mixing in that setting and questions about other programs and finite groups.

Related work. Black-box obfuscation is impossible for general circuits, as shown by Barak, Goldreich, Impagliazzo, Rudich, Sahai, Vadhan, and Yang [BGI⁺01]; this motivates the weaker notion of indistinguishability obfuscation. The first iO candidate from branching programs [GGH⁺13] was followed by constructions for specific function classes: lockable obfuscation for point functions from LWE [GKW17, WZ17], conjunction obfuscation from noisy encodings and LPN [BKM⁺18, BLMZ19], and entropic Ring-LWE approaches [BVWW16]. The landmark result of Jain, Lin, and Sahai [JLS21] achieved iO for general circuits from well-founded assumptions, departing from the branching-program paradigm. Recent lattice-based candidates [BDGM20, WW21, HJL25] depart from the S_5 branching-program paradigm and rest on lattice assumptions, while impossibility results [DJM⁺25] constrain the space of viable assumptions. Mixed-input evaluation — running the program on inputs that read a variable inconsistently — is the setting in which the algebraic behaviour of a branching-program layer is analysed [CVW18], and is the setting of the experiment we study. Annihilation attacks [MSZ16] demonstrated concrete vulnerabilities in early branching-program obfuscation candidates, underscoring the importance of understanding the algebraic structure of inconsistent evaluations.

Random subproducts. The unconditional law P_T studied here is a *random subproduct*: a fixed word $g_1^{\varepsilon_1} \cdots g_L^{\varepsilon_L}$ in which each prescribed group element is independently activated, $\varepsilon_i \sim \text{Ber}(1/2)$. Such products have been studied as lazy-transposition shuffles and, more generally, under the name of group mixability [AH18, ABGV25]; those works ask whether one can *choose* a sequence—typically together with its activation probabilities—whose product is exactly uniform, and show in particular that symmetric groups are mixable. Our setting differs in both respects: every activation probability is fixed to $1/2$, and the word is not free but is the recursive expansion of a valid balanced Barrington commutator template. Exact uniformity is then impossible for P_T , since each of its atoms has dyadic probability while $1/|S_5| = 1/120$ is not dyadic. We instead prove an approximate-mixing bound, uniform over this constrained family, and transfer it to the conditioned law D_T (Lemma 1).

Fourier-analytic methods for branching programs have a separate tradition in the pseudorandomness literature [RSV13, LPV22], but those works use Fourier analysis on $\{0, 1\}^n$ (the input space) to fool Boolean outputs of read-once programs. Our analysis instead uses Fourier analysis on S_5 (the output group) to bound total variation of the group-valued output distribution of multi-read programs — a fundamentally different analytic

object requiring different tools. Canetti, Chamon, Mucciolo, and Ruckenstein [CCMR24] recently proposed a “local mixing” framework for general-purpose obfuscation based on reversible circuits and thermodynamic arguments; their mixing notion is structural (circuit-level), whereas ours is distributional (output-group-level), making the two approaches complementary. Our analysis applies the Diaconis–Shahshahani framework [DS81] to the output group of the program.

Techniques. The proof of T-independence is purely combinatorial. The proof of spectral contraction uses three different methods at three different scales: exact symbolic algebra (levels 0–2), analytic structure and exact integer certification (level 3), with independent exhaustive numerical cross-checks, and analytic induction (levels 4+). The key discovery that bridges levels 2 and 3 is the simplex structure of the preserved directions: in the reduced level-2 census, the 270 out of 405 configurations with operator norm 1 each preserve one of exactly 5 directions in $\mathbb{R}^4$, forming a regular simplex with pairwise $|\cos \theta| = 1/4$. An algebraic proof shows that valid level-3 configurations cannot share a preserved direction: within the stabilizer $\text{Stab}(k) \cong S_4$, the derived series terminates at depth 3, forcing the outer commutator to be trivial. A separate analysis of S_n for $n \geq 6$ shows that this mechanism is specific to S_5 : the level-1 Fourier transform preserves a subspace of dimension $n - 3$ in $\mathbb{R}^{n-1}$, and the intersection bound $2(n - 3) - (n - 1) = n - 5 \geq 1$ forces $\|\widehat{D}_2\|_{\text{op}} = 1$ for every level-2 quadruple. At level 3, explicit norm-1 configurations rule out uniform contraction for S_6 and S_7 , whereas a further dimensional bound forces every configuration to have norm 1 for $n \geq 10$. The analysis here does not decide $n \in \{8, 9\}$.

2 Preliminaries

2.1 Branching Programs and Barrington’s Theorem

Definition 1 (Permutation Branching Program). A *permutation branching program* of width w and length L over a group $G \leq S_w$ is a sequence of instructions $\Pi = ((i_1, g_1^{(0)}, g_1^{(1)}), \dots, (i_L, g_L^{(0)}, g_L^{(1)}))$, where each $i_j \in [\ell]$ is a variable index and $g_j^{(0)}, g_j^{(1)} \in G$. On input $x \in \{0, 1\}^\ell$, the program evaluates $\Pi(x) = \prod_{j=1}^L g_j^{(x_{i_j})}$. The program α -computes a function f if $\Pi(x) = \alpha$ when $f(x) = 1$ and $\Pi(x) = e$ when $f(x) = 0$.

Theorem 1 (Barrington [Bar89]). *Every Boolean function computed by a circuit of depth d can be α -computed by a permutation branching program of width 5 and length at most 4^d over S_5 , for some $\alpha \neq e$.*

The construction uses the *commutator trick*: given programs P and Q that α -compute f and β -compute g respectively, the concatenation $P \cdot Q \cdot P^{-1} \cdot Q^{-1}$ computes $f \wedge g$ via the group commutator $[\alpha, \beta] = \alpha\beta\alpha^{-1}\beta^{-1}$. This works because S_5 contains enough non-commuting elements to choose α, β and their conjugates so that the required commutators remain non-identity at each gate. Non-identity elements can of course commute; the Barrington construction depends on these careful choices rather than on an arbitrary commutator being nontrivial.

Definition 2 (Valid balanced transposition template). A *balanced transposition template* $\mathcal{T}$ of level r assigns a transposition of S_5 to each leaf of the complete binary tree of depth r , together with a fixed orientation at each node. Its *target* is defined recursively: a level-0 configuration is a transposition, with target itself; a level- $(r+1)$ configuration is an ordered pair (A, B) of level- r configurations, with target $\alpha_{(A,B)} = [\alpha_A, \alpha_B]$. The template is *valid* if every configuration in it—each sub-block and the root—has non-identity target; equivalently, $\alpha_A \neq e$, $\alpha_B \neq e$ and $[\alpha_A, \alpha_B] \neq e$ at every node. We refer to the valid

configurations of a given level as valid *pairs* (level 1), *quadruples* (level 2) and *level-3 configurations*.

All point-function mixing bounds below are proved *for every fixed valid balanced transposition template*: a template $\mathcal{T}$ is fixed first (it is what the comparison of two targets T and T' holds constant), and the bounds then hold uniformly over all valid templates. The notation D_T suppresses the dependence on $\mathcal{T}$, which is fixed throughout. (Statements of wider scope— T -independence for a common instruction template, conjunctions, and other groups—are stated with their own hypotheses where they appear.)

Non-vacuity. Valid templates exist at every level $r \geq 3$. On the points $\{0, 1, 2, 3, 4\}$, the level-3 tree

$$\mathcal{T}_3 = \left[\left[[(01), (02)], [(01), (04)] \right], \left[[(01), (02)], [(01), (13)] \right] \right]$$

is valid and has target $\rho = (01234)$. Moreover, $[(02431), (02314)] = \rho$, and both inputs are conjugate to ρ . Simultaneously conjugating all leaves of a valid template by a fixed group element preserves transposition leaves and validity while conjugating every internal target. Thus two suitably conjugated copies of a level- r template with target ρ form a valid level- $(r+1)$ template with the same target; iteration proves the claim.

Definition 3 (Point Function). The point function $\text{CC}_T : \{0, 1\}^\ell \rightarrow \{0, 1\}$ is defined by $\text{CC}_T(x) = 1$ if and only if $x = T$.

For point functions, we restrict to the following fixed balanced transposition-template construction. The template has ℓ leaves (one per bit position), each comparing x_i to T_i . Leaf i outputs a fixed transposition σ_i when $x_i = T_i$ and the identity e when $x_i \neq T_i$. The leaves are combined by a fixed balanced binary tree of commutators, with a fixed orientation at every internal node. When ℓ is a power of two, the tree has depth $\log_2 \ell$, yielding a program of length $L = 4^{\log_2 \ell} = \ell^2$. All statements below concern this template and the branch swaps induced by changing T . They do not claim uniform mixing for arbitrary width-5 Barrington programs or for templates using only even permutations such as 5-cycles.

2.2 The Distribution D_T

Definition 4 (Inconsistent Path). A bit string $b = (b_1, \dots, b_L) \in \{0, 1\}^L$ is *inconsistent* with respect to the program Π if there exist steps j, j' reading the same variable ($i_j = i_{j'}$) with $b_j \neq b_{j'}$.

Definition 5 (D_T). Let $\mathcal{I} \subset \{0, 1\}^L$ be the set of inconsistent paths. The instructions $g_j^{(\cdot)}$ are those of the program BP_T for the point function CC_T ; the subscript T records this dependence (by Theorem 3, D_T is in fact independent of T). The distribution D_T over S_5 is defined by

$$D_T(\pi) = \Pr_{b \sim \text{Uniform}(\mathcal{I})} \left[\prod_{j=1}^L g_j^{(b_j)} = \pi \right].$$

2.3 Representation Theory of S_5

The symmetric group S_5 has 7 irreducible representations, labeled by partitions of 5:

Partition λ	Dimension d_λ	$\chi_\lambda(\text{transposition})$	Name
(5)	1	1	Trivial
(4, 1)	4	2	Standard
(3, 2)	5	1	—
(3, 1, 1)	6	0	—
(2, 2, 1)	5	−1	Sign $\otimes$ (3, 2)
(2, 1, 1, 1)	4	−2	Sign $\otimes$ Standard
(1 ⁵)	1	−1	Sign

The dimensional check confirms: $1^2 + 4^2 + 5^2 + 6^2 + 5^2 + 4^2 + 1^2 = 120 = |S_5|$.

We call two transpositions $\sigma, \tau \in S_5$ a *non-commuting pair* if $[\sigma, \tau] \neq e$, equivalently if they share exactly one element. There are 30 such unordered pairs among the 10 transpositions of S_5 .

For a matrix $M \in \mathbb{R}^{d \times d}$, we write $\|M\|_F^2 = \text{tr}(M^\top M) = \sum_{i,j} M_{ij}^2$ for the squared Frobenius norm and $\|M\|_{\text{op}} = \max_{\|v\|=1} \|Mv\|$ for the operator norm (largest singular value). The key inequality relating them is $\|M\|_{\text{op}}^2 \leq \|M\|_F^2 \leq d \cdot \|M\|_{\text{op}}^2$.

Definition 6 (Standard Representation). The standard representation $\rho_{(4,1)} : S_5 \rightarrow \text{GL}(\mathbb{R}^4)$ acts on the hyperplane $V = \{v \in \mathbb{R}^5 : \sum_i v_i = 0\}$ by permuting coordinates. We use the orthonormal basis

$$\begin{aligned} f_1 &= (e_0 - e_1)/\sqrt{2}, & f_2 &= (e_0 + e_1 - 2e_2)/\sqrt{6}, \\ f_3 &= (e_0 + e_1 + e_2 - 3e_3)/\sqrt{12}, & f_4 &= (e_0 + e_1 + e_2 + e_3 - 4e_4)/\sqrt{20}, \end{aligned}$$

so that $\rho_{(4,1)}(\sigma)$ is orthogonal for all $\sigma \in S_5$.

2.4 The Diaconis–Shahshahani Upper Bound Lemma

Theorem 2 (Diaconis–Shahshahani [DS81]). *For any probability distribution Q on a finite group G ,*

$$4 \cdot d_{\text{TV}}(Q, U)^2 \leq \sum_{\rho \neq \text{trivial}} d_\rho \cdot \|\hat{Q}(\rho)\|_F^2,$$

where the sum is over non-trivial irreducible representations ρ of dimension d_ρ , $\hat{Q}(\rho) = \sum_{g \in G} Q(g) \rho(g)$ is the Fourier transform, and $\|\cdot\|_F$ denotes the Frobenius norm.

For S_5 , the standard representation (4, 1) is the *mixing bottleneck*. Although (2, 1, 1, 1) = sign $\otimes$ (4, 1) shares the same normalized character ratio $|\chi(\sigma)|/d_\rho = 1/2$, its leaf projection $\frac{1}{2}(I - \rho_{(4,1)}(\sigma))$ has rank 1 (versus rank 3 for (4, 1)), causing it to contract at level 1 with operator norm $\leq (1/2)^3 = 1/8$. The remaining representations satisfy $|\chi(\sigma)|/d_\rho \leq 1/5$ and likewise contract earlier (Proposition 1). We therefore focus the exposition on $\|\hat{D}(4, 1)\|_F^2$, while Proposition 1 and the level-3 certificate control every other non-trivial irreducible, as the Diaconis–Shahshahani bound requires.

3 Exact T-Independence

We first show that the output distribution carries no information about the target: for programs built from a common template, D_T is literally independent of T . The argument is a short combinatorial bijection that uses no property of S_5 , so we give it before the group-specific mixing analysis.

Theorem 3 (T-Independence). *Let BP_T be the Barrington branching program for the point function CC_T over $\{0, 1\}^\ell$, with all programs built from the same instruction template:*

the same variable-read sequence and the same transposition at each leaf, with the target T only determining which branch receives that transposition. Then $D_T = D_{T'}$ for all $T, T' \in \{0, 1\}^\ell$.

Proof. Let $S = \{i : T_i \neq T'_i\}$ be the set of differing bit positions. Define $F_S : \{0, 1\}^L \rightarrow \{0, 1\}^L$ by

$$F_S(b)_j = \begin{cases} 1 - b_j & \text{if } i_j \in S, \\ b_j & \text{otherwise,} \end{cases}$$

where i_j is the variable read at step j .

We verify three properties:

(1) F_S is an involution on $\{0, 1\}^L$. Applying F_S twice flips each bit in S -positions twice, returning to the original string. Hence $F_S \circ F_S = \text{Id}$, and in particular F_S is a bijection on the full space $\{0, 1\}^L$.

(2) F_S preserves inconsistency. F_S flips all readings of each variable x_i with $i \in S$ simultaneously. If all readings of x_i were equal (say, all 0), after flipping they are all 1 — still consistent. If some readings differed, after flipping they still differ. Variables with $i \notin S$ are unaffected. Therefore inconsistency of each variable is preserved individually.

(3) $\varphi_{T'}(b) = \varphi_T(F_S(b))$ for all b . At each step j reading variable x_{i_j} :

- If $i_j \notin S$: $T_{i_j} = T'_{i_j}$ and F_S does not flip b_j . Same input, same target bit, same output.
- If $i_j \in S$: $T'_{i_j} = 1 - T_{i_j}$ and F_S flips b_j . The swap of target bit and input bit cancel: $g_j^{(b_j)}$ under T' equals $g_j^{(1-b_j)}$ under T , which equals $g_j^{(F_S(b)_j)}$ under T .

Hence the per-step outputs, and their product, are equal.

Combining (1)–(3):

$$\begin{aligned} D_{T'}(\pi) &= \Pr[\varphi_{T'}(b) = \pi \mid b \in \mathcal{I}] \\ &= \Pr[\varphi_T(F_S(b)) = \pi \mid b \in \mathcal{I}] && \text{(by (3))} \\ &= \Pr[\varphi_T(b') = \pi \mid F_S^{-1}(b') \in \mathcal{I}] && \text{(substituting } b' = F_S(b)) \\ &= \Pr[\varphi_T(b') = \pi \mid b' \in \mathcal{I}] && \text{(by (2), since } F_S = F_S^{-1} \text{ by (1))} \\ &= D_T(\pi). && \square \end{aligned}$$

Remark 1. The proof uses no property of S_5 — it works for branching programs of any width over any group, provided the programs for different targets share the same instruction template and differ only in branch assignment at the literals whose target bit changes. Thus the argument is not specific to Barrington’s particular choice of transpositions, but it is a template-level statement rather than a statement about arbitrary independently chosen programs.

Remark 2. The proof extends to conjunctions with wildcards: if position i is a wildcard, the corresponding leaf outputs the same permutation σ_i for both bit values. Flipping b_j at wildcard positions changes both branches identically, so the output is unchanged. The bijection F_S is restricted to non-wildcard differing positions.

Remark 3 (Unconditional T-independence). The bijection F_S preserves inconsistency of each variable individually (property (2)), and therefore also preserves the set of *consistent* paths. Consequently, $D_T = D_{T'}$ holds not only for the distribution conditioned on inconsistent paths, but also for the unconditional distribution over all 2^L paths and for the distribution over consistent paths alone. The conditioning on inconsistency is what makes the mixing statement meaningful — on consistent paths the output is determined by the input, and the accepting input is T itself — but it is not needed for T-independence, which holds for all three laws.

Remark 4 (Template dependence). The flip bijection compares targets inside a fixed instruction template: the same variable-read sequence, the same leaf transpositions, and the same commutator tree. It should not be read as an invariance statement over different templates. Changing the grouping of leaves or the orientation of internal commutators can change the conditioned distribution, even when the resulting program computes the same point function. The spectral results below are therefore stated for the fixed balanced template.

4 Spectral Contraction

Having established that D_T is T -independent, we now prove the harder property: that D_T is negligibly close to uniform on S_5 . This is where the representation theory of S_5 enters, and the crux is a level-2-to-level-3 step in which a hidden simplex geometry forces contraction. Since D_T is independent of T (Theorem 3), it suffices to show $d_{\text{TV}}(D_T, U) \leq \text{negl}(\ell)$ for any fixed T ; by the Diaconis–Shahshahani bound (Theorem 2), this reduces to bounding $\|\hat{D}(\rho)\|_{\text{F}}^2$ for each non-trivial representation ρ .

Lemma 1 (Conditioning). *The Fourier analysis below operates on the unconditional distribution P_T in which each of the L bits is drawn independently and uniformly; its Fourier transform factorizes as a product of per-leaf projection operators P_σ . Let $p = \Pr[b \text{ consistent}]$ and let C_T denote the distribution conditioned on consistent paths, so that $P_T = (1 - p) D_T + p C_T$. Then*

$$d_{\text{TV}}(D_T, U) \leq d_{\text{TV}}(P_T, U) + p d_{\text{TV}}(D_T, C_T) \leq d_{\text{TV}}(P_T, U) + p.$$

For the point-function program every variable is read at least once, so the consistent paths are in bijection with $\{0, 1\}^\ell$; hence $p = 2^\ell / 2^L = 2^{-\ell(\ell-1)}$ (using $L = \ell^2$), which is negligible.

Proof. From $P_T = (1 - p) D_T + p C_T$ we get $P_T - D_T = p(C_T - D_T)$, so $d_{\text{TV}}(D_T, P_T) = p d_{\text{TV}}(D_T, C_T) \leq p$. The triangle inequality $d_{\text{TV}}(D_T, U) \leq d_{\text{TV}}(D_T, P_T) + d_{\text{TV}}(P_T, U)$ then gives the bound. The count $|\{b : b \text{ consistent}\}| = 2^\ell$ holds because each of the ℓ variables is read at least once, so assigning a single value to each variable is a bijection onto the consistent paths. $\square$

Remark 5 (Sign representation vanishes identically). For the sign representation (1^5), the leaf Fourier transform is $\hat{D}(\text{sign}) = \frac{1}{2}(\text{sign}(\sigma) + 1) = 0$ for any transposition σ , since $\text{sign}(\sigma) = -1$. This propagates to all levels by multiplicativity, so the sign representation contributes nothing to the Diaconis–Shahshahani bound. The sum thus runs over 5 (not 6) non-trivial representations.

Proposition 1 (Contraction in the remaining representations). *For all five non-trivial, non-sign representations ρ of S_5 , the operator norm $\|\hat{D}_3(\rho)\|_{\text{op}}$ is strictly less than 1 for every valid level-3 configuration. The representation $(4, 1)$ is the unique bottleneck:*

ρ	d_ρ	Leaf rank	Numerical $\max \ \hat{D}_3(\rho)\ _{\text{op}}$	First level < 1
$(4, 1)$	4	3	3.54×10^{-2}	Level 3
$(3, 2)$	5	3	1.93×10^{-5}	Level 2
$(3, 1, 1)$	6	3	1.05×10^{-5}	Level 2
$(2, 2, 1)$	5	2	$< 10^{-10}$	Level 1
$(2, 1, 1, 1)$	4	1	$< 10^{-10}$	Level 1

The representations $(2, 1, 1, 1) = \text{sign} \otimes (4, 1)$ and $(2, 2, 1) = \text{sign} \otimes (3, 2)$ have leaf projections of rank 1 and 2 respectively. For rank-1 projections $P_\sigma = v_\sigma v_\sigma^\top$, the level-1

operator $(P_\sigma P_\tau)^2 = \langle v_\sigma, v_\tau \rangle^3 v_\sigma v_\tau^\top$, giving $\|(P_\sigma P_\tau)^2\|_{\text{op}} = |\langle v_\sigma, v_\tau \rangle|^3 = (1/2)^3 = 1/8$ for non-commuting pairs. So these representations contract immediately at level 1. The representations $(3, 2)$ and $(3, 1, 1)$ have rank-3 leaf projections like $(4, 1)$, but lack the simplex obstruction (Observation 6): all 405 level-2 quadruples have operator norm < 0.085 , so they contract one level earlier. The displayed values are independently reproduced in floating point over the full ordered level-3 universe (Script 18). The uniform bound used in the paper is certified separately in exact integer arithmetic for all five representations (Script 20; Section 7).

We focus on the bottleneck representation $(4, 1)$.

4.1 Framework: Projections and the Fourier Domain

For a transposition $\sigma = (a\ b) \in S_5$, the standard representation $\rho_{(4,1)}(\sigma)$ has eigenvalues $\{+1, +1, +1, -1\}$ (three fixed directions and one flipped). The Fourier transform of the leaf distribution $\{\sigma : 1/2, e : 1/2\}$ is

$$P_\sigma := \frac{1}{2}(\rho_{(4,1)}(\sigma) + I_4),$$

which is a rank-3 orthogonal projection (idempotent, symmetric, trace 3).

Lemma 2. *For any transposition $\sigma \in S_5$:*

- (a) $P_\sigma^2 = P_\sigma$ (idempotent),
- (b) $P_\sigma^\top = P_\sigma$ (symmetric),
- (c) $\text{rank}(P_\sigma) = 3$,
- (d) $\|P_\sigma\|_{\mathbb{F}}^2 = \text{tr}(P_\sigma) = 3$.

The null space of P_σ is spanned by the projection of $e_a - e_b$ onto the hyperplane V , where $\sigma = (a\ b)$.

Throughout this section, $\widehat{D}_r$ denotes the Fourier operator of a level- r configuration of the *unconditional* law P_T (independent uniform bits at every step); it is not the Fourier transform of the conditioned law D_T , which enters only through the conditioning lemma (Lemma 1). At each level of the Barrington commutator tree, this operator is a product of projections. For the single AND block at level 1 using transpositions σ, τ (with $\sigma^{-1} = \sigma$, $\tau^{-1} = \tau$ since they are transpositions):

$$\widehat{D}_1 = P_\sigma \cdot P_\tau \cdot P_\sigma \cdot P_\tau = (P_\sigma P_\tau)^2.$$

At level 2, using two level-1 blocks A (with σ_1, τ_1) and B (with σ_2, τ_2):

$$\widehat{D}_2 = (P_{\sigma_1} P_{\tau_1})^2 \cdot (P_{\sigma_2} P_{\tau_2})^2 \cdot (P_{\tau_1} P_{\sigma_1})^2 \cdot (P_{\tau_2} P_{\sigma_2})^2,$$

where the inverse blocks reverse the transposition order.

4.2 Level 0 to Level 1 Contraction

Theorem 4 (Universal Level-1 Contraction). *For any non-commuting pair of transpositions (σ, τ) in S_5 :*

$$\|(P_\sigma P_\tau)^2\|_{\mathbb{F}}^2 = \frac{129}{64}.$$

The contraction ratio is $\frac{129/64}{3} = \frac{43}{64} \approx 0.6719$, and is **universal** — identical for all 30 non-commuting transposition pairs.

Proof. There are $\binom{5}{2} = 10$ transpositions in S_5 , forming 30 non-commuting pairs (sharing exactly one element). For each pair, we compute $(P_\sigma P_\tau)^2$ in the orthonormal basis of Section 2 using exact symbolic arithmetic over algebraic numbers (the basis vectors carry radicals, which are represented exactly), and verify $\|(P_\sigma P_\tau)^2\|_F^2 = 129/64$.

The universality follows from the geometry of the standard representation: the null vectors of P_σ and P_τ (projections of $e_a - e_b$ and $e_c - e_d$ onto V) have inner product $\pm 1/2$ for all non-commuting pairs, determined by the simplex geometry of $\{e_0, \dots, e_4\}$ in V . Since $\|(P_\sigma P_\tau)^2\|_F^2$ depends only on this angle, it is universal. $\square$

4.3 Level 1 to Level 2 Contraction

Theorem 5 (Level-2 Contraction). *For any valid quadruple of transpositions $(\sigma_1, \tau_1, \sigma_2, \tau_2)$ with $[\sigma_i, \tau_i] \neq e$ and $[[\sigma_1, \tau_1], [\sigma_2, \tau_2]] \neq e$:*

$$\|\widehat{D}_2\|_F^2 \leq \frac{67586345}{67108864} \approx 1.0071.$$

The contraction ratio relative to level 1 is at most $\frac{67586345/67108864}{129/64} \approx 0.4997 < 1$.

Proof. We first enumerate the reduced orientation quotient: 405 valid quadruple classes from the 30 non-commuting unordered level-1 pairs, with the additional constraint that the outer commutator is nontrivial. For each class, we compute $\widehat{D}_2$ in exact symbolic arithmetic over algebraic numbers and evaluate $\|\widehat{D}_2\|_F^2$. The reduced computation yields 16 distinct Frobenius norms, ranging from $82431849/1073741824 \approx 0.077$ to the worst case $67586345/67108864$.

Because the theorem is stated for ordered quadruples, we also run the ordered level-2 exact cross-check (Script 19): 60 ordered level-1 pairs give 3240 ordered valid level-2 blocks. This larger ordered universe has 17 distinct Frobenius norms (one additional non-maximal value), but the same maximum $67586345/67108864$. $\square$

4.4 The Simplex Obstruction

Despite the Frobenius-norm contraction at level 2, the operator norm $\|\widehat{D}_2\|_{\text{op}}$ equals 1 for 270 of the 405 valid quadruples in the reduced census, blocking a naive submultiplicativity argument for higher levels. The following observation explains the structure of this obstruction.

Observation 6 (Simplex Structure). *In the reduced census, the 270 quadruples with $\|\widehat{D}_2\|_{\text{op}} = 1$ each preserve exactly one direction in $\mathbb{R}^4$ (as the eigenvector of $\widehat{D}_2^\top \widehat{D}_2$ with eigenvalue 1). These directions are precisely the projections of the 5 standard basis vectors $e_0, \dots, e_4 \in \mathbb{R}^5$ onto the hyperplane $V = \{\sum v_i = 0\}$:*

$$v_k = \frac{e_k - \bar{e}}{\|e_k - \bar{e}\|}, \quad \bar{e} = \frac{1}{5}(1, 1, 1, 1)^\top, \quad k = 0, \dots, 4.$$

*These 5 directions form a **regular simplex** in $\mathbb{R}^4$, with pairwise inner products $|\langle v_i, v_j \rangle| = 1/4$ for all $i \neq j$. Each direction is preserved by exactly 54 quadruples (perfectly symmetric).*

Observation 6 exhausts this reduced census. The following lemma supplies the structural implication that the level-3 argument actually uses: preserving v_k forces every leaf of the block to fix the point k .

Lemma 3 (Stabilizer characterization of preserved directions). *Let B be a valid level-2 block with leaf transpositions $\sigma_1, \tau_1, \sigma_2, \tau_2$, and let $\widehat{D}_2(B)$ be its level-2 operator in the standard representation. For each $k \in \{0, \dots, 4\}$:*

$$\|\widehat{D}_2(B) v_k\| = \|v_k\| \quad \Longleftrightarrow \quad \sigma_i(k) = \tau_i(k) = k \quad (i = 1, 2),$$

and in that case $\widehat{D}_2(B)v_k = v_k$. Equivalently: a level-2 block preserves the direction v_k if and only if all four of its leaf transpositions lie in the point stabilizer $\text{Stab}(k) \cong S_4$.

Proof. Write $\widehat{D}_2(B) = P_{t_1}P_{t_2} \cdots P_{t_r}$ as the product of the $r = 16$ leaf projections appearing in its expansion (each factor is $P_t = \frac{1}{2}(\rho_{(4,1)}(t) + I)$ for one of the four leaf transpositions t ; the inverse sub-blocks contribute the same projections in reversed order, since each transposition is an involution). By Lemma 2 every P_t is an orthogonal projection, hence $\|P_t w\| \leq \|w\|$ for all w .

($\Rightarrow$) Suppose $\|\widehat{D}_2(B)v_k\| = \|v_k\|$. Setting $w_0 = v_k$ and $w_j = P_{t_{r-j+1}}w_{j-1}$, the chain

$$\|v_k\| = \|w_r\| \leq \|w_{r-1}\| \leq \cdots \leq \|w_0\| = \|v_k\|$$

collapses to equality at every step. For an orthogonal projection, $\|Pw\| = \|w\|$ forces $Pw = w$ (indeed $\|w\|^2 = \|Pw\|^2 + \|(I - P)w\|^2$, so $\|(I - P)w\| = 0$). Applying this successively gives $P_t v_k = v_k$ for every leaf transposition t occurring in the product.

Now $P_t v_k = v_k$ means $\rho_{(4,1)}(t)v_k = v_k$. Since v_k is the projection of the basis vector e_k onto the sum-zero hyperplane V and $\rho_{(4,1)}$ is the restriction of the permutation action to V , we have $\rho_{(4,1)}(t)v_k = v_{t(k)}$. The five vectors $v_0, \dots, v_4$ are pairwise distinct (they satisfy $\langle v_i, v_j \rangle = -1/4 \neq 1$ for $i \neq j$), so $v_{t(k)} = v_k$ forces $t(k) = k$. Hence every leaf transposition of B fixes k , i.e. all four lie in $\text{Stab}(k)$.

($\Leftarrow$) If all four leaf transpositions fix k then $\rho_{(4,1)}(t)v_k = v_{t(k)} = v_k$ for each of them, so $P_t v_k = v_k$ and therefore $\widehat{D}_2(B)v_k = v_k$, which has the same norm as v_k . $\square$

The regularity of this structure — five directions, uniform distribution, constant pairwise angle — might suggest a general phenomenon. The following theorem shows the opposite: the simplex structure is specific to S_5 , and for S_n with $n \geq 6$ a dimensional obstruction prevents the same level-2 direction-mixing argument from proving a uniform level-3 gap.

Theorem 7 (Dimensional obstruction for S_n , $n \geq 6$). *For S_n with $n \geq 5$, the level-1 Fourier transform $\widehat{D}_1 = (P_\sigma P_\tau)^2$ in the standard representation $(n-1, 1)$ has singular values $\underbrace{1, \dots, 1}_{n-3}, 1/8, 0$. In particular, $\widehat{D}_1$ preserves a subspace of dimension $n-3$ isometrically.*

For $n \geq 6$, the minimum intersection of two such preserved subspaces in $\mathbb{R}^{n-1}$ is $2(n-3) - (n-1) = n-5 \geq 1$, so every valid level-2 quadruple has $\|\widehat{D}_2\|_{\text{op}} = 1$ and the simplex-breaking mechanism of Lemma 4 does not extend by the same argument. At level 3, the maximum operator norm is 1 for S_6 and S_7 , whereas every configuration has norm 1 for $n \geq 10$. Thus S_5 is the only case with uniform level-3 contraction among the cases settled here; the present analysis does not decide $n \in \{8, 9\}$.

Proof. The projections $P_\sigma = I_V - v_\sigma v_\sigma^\top$ and $P_\tau = I_V - v_\tau v_\tau^\top$ (where v_σ, v_τ are the unit null vectors of the transpositions in V) share the property that $\langle v_\sigma, v_\tau \rangle = -1/2$, after choosing their signs, for all non-commuting pairs (independent of n). Decomposing $V = W \oplus W^\perp$ where $W = \text{span}(v_\sigma, v_\tau)$ has dimension 2: on $W^\perp$ (dimension $n-3$), both projections act as the identity, so $\widehat{D}_1 = (P_\sigma P_\tau)^2 = I$ on $W^\perp$, contributing $n-3$ singular values equal to 1. On W , the restriction is a 2×2 block independent of n ; a direct computation gives $\widehat{D}_1^\top \widehat{D}_1|_W$ with eigenvalues $1/64$ and 0 (trace = $4/256$, determinant = 0), so the singular values on W are $1/8$ and 0.

For $n \geq 6$: two subspaces of dimension $n-3$ in $\mathbb{R}^{n-1}$ intersect in dimension $\geq 2(n-3) - (n-1) = n-5 \geq 1$. Any unit vector in this intersection is preserved isometrically by both level-1 sub-blocks of a level-2 commutator, so $\|\widehat{D}_2\|_{\text{op}} = 1$ for every valid quadruple.

The same counting one step up controls level 3. Each level-2 operator with $\|\widehat{D}_2\|_{\text{op}} = 1$ fixes a subspace of dimension $\geq n-5$, and two such subspaces in $\mathbb{R}^{n-1}$ share a common fixed vector once $2(n-5) - (n-1) = n-9 \geq 1$, i.e. for $n \geq 10$; then $\|\widehat{D}_3\|_{\text{op}} = 1$ as

well. The value $n = 9$ is exactly where disjoint placement becomes possible again, so the dimensional argument alone does not settle $n \in \{8, 9\}$. For the two immediate neighbours, the depth-3 seed given after Definition 2, viewed in S_6 or S_7 , remains valid and fixes the last point. Every leaf projection therefore fixes the nonzero vector $(1, \dots, 1, -(n-1)) \in V$, so its level-3 operator has norm 1. Script 14 also checks the maximum numerically over the reduced census of 1,242,915 configurations in S_6 and 10,282,335 in S_7 . This is a statement about the maximum, not every configuration: the script includes exact Gram-form certificates for strictly contractive examples in both groups. Thus uniform level-3 contraction fails in these two cases. $\square$

Remark 6. The one-dimensional simplex mechanism of Observation 6 does not generalize cleanly to S_n for $n \geq 6$: exhaustive computation confirms that all 1620 quadruples in the reduced census for S_6 and all 4725 for S_7 have operator norm 1 (versus $270/405 = 66.7\%$ for S_5). In these larger groups the preserved isometric subspaces have forced positive-dimensional intersections rather than the disjoint direction placement that drives the S_5 argument. Theorem 7 explains this: S_5 is the unique case where the preserved subspace dimension ($n - 3 = 2$) is small enough relative to the ambient dimension ($n - 1 = 4$) to permit disjoint placement. The theorem is verified by two independent methods, including exact symbolic arithmetic over all adjacent pairs in S_5, S_6, S_7 and exhaustive enumeration of the reduced level-2 censuses (Scripts 14 and 15).

An immediate consequence is that the level-3 spectral contraction mechanism presented here does not directly extend to the width > 5 programs that appear in optimized obfuscation constructions avoiding Barrington’s 4^d blowup [AGIS14]. Establishing unconditional mixing for such programs requires fundamentally different techniques, which we discuss in Section 6.

4.5 Level 2 to Level 3 Contraction

Why switch to the operator norm. The Frobenius norm tracks total “energy” of the Fourier operator and is computed exactly at the first two levels (Theorems 4–5); the induction from level 3 onward uses the operator norm. However, the worst-case $\|\hat{D}_2\|_F^2 \approx 1.007 > 1$, so the absolute Frobenius norm has not yet dropped below 1, and an inductive argument based on Frobenius norms alone does not yield $\|\hat{D}_d\|_F \rightarrow 0$. The operator norm resolves this: once $\|\hat{D}_3\|_{\text{op}} < 1$, submultiplicativity raises the uniform level- d bound to its fourth power, which iterates to doubly-exponential decay in the depth. The cost is that we need operator-norm contraction at level 3 for *every* configuration, not just worst-case Frobenius contraction.

Lemma 4 (Direction Mixing). *For every valid level-3 configuration $(\text{quad}_A, \text{quad}_B)$ in which both sub-blocks have $\|\hat{D}_2\|_{\text{op}} = 1$, the two sub-blocks preserve **different** directions. In the reduced census, there are 29160 such pairs, none sharing the same preserved direction.*

Proof. Suppose for contradiction that both quad_A and quad_B preserve the same direction v_k (the projection of e_k onto V). By Lemma 3, a quadruple preserving v_k consists of transpositions that fix element k , so all four transpositions in each sub-block lie in the stabilizer $\text{Stab}(k) \cong S_4$.

The derived series of S_4 is $S_4 \triangleright A_4 \triangleright V_4 \triangleright \{e\}$, where $V_4 = \{e, (01)(23), (02)(13), (03)(12)\}$ is the Klein four-group (abelian). In the Barrington commutator tree:

- Level-1 targets $[\sigma_i, \tau_i]$ are commutators of transpositions in S_4 , hence lie in $[S_4, S_4] = A_4$.
- Level-2 targets $[[\sigma_1, \tau_1], [\sigma_2, \tau_2]]$ are commutators of elements of A_4 , hence lie in $[A_4, A_4] = V_4$.

Since V_4 is abelian, the outer commutator $[\alpha_A, \alpha_B] = e$ for any $\alpha_A, \alpha_B \in V_4$. This contradicts the requirement $[\alpha_A, \alpha_B] \neq e$ for a valid level-3 configuration.

The result is also confirmed by exhaustive enumeration over the reduced configuration set (Script 11). $\square$

Theorem 8 (Level-3 Contraction). *For every valid level-3 configuration:*

$$\|\hat{D}_3\|_{\text{op}} \leq \lambda := 0.0355.$$

In particular, $\|\hat{D}_3\|_{\text{op}} < 1$ for all configurations in the fixed transposition template.

Proof. For each valid level-3 pair $(\text{quad}_A, \text{quad}_B)$, we compute

$$\hat{D}_3 = \hat{D}_{2,A} \cdot \hat{D}_{2,B} \cdot \hat{D}_{2,A}^\top \cdot \hat{D}_{2,B}^\top$$

over the full ordered universe: all five non-trivial, non-sign irreducible representations over the 3,240 ordered level-2 blocks, i.e. 9,447,840 oriented pairs. Script 20 reduces this universe, with exact orbit accounting, to 39,591 representatives under simultaneous S_5 -conjugation and exchange of the two blocks.

The five relevant irreducibles are carried by four integral carrier modules derived from permutation actions (with M_9^s obtained by a sign twist), so that no irrational basis is ever needed. Let M_{41} be the sum-zero submodule of the natural action on the five points, which realises $(4, 1)$; let M_9 be the sum-zero submodule of the action on the ten 2-subsets, whose decomposition is $(4, 1) \oplus (3, 2)$; let M_9^s be the sign twist of M_9 , carrying $(2, 1, 1, 1) \oplus (2, 2, 1)$; and let $M_6 = \bigwedge^2 M_{41}$, which realises $(3, 1, 1)$. The invariant Gram forms are $G = I + J$ on M_{41} , M_9 , and M_9^s (with J the all-ones matrix on the ambient coordinates) and $\bigwedge^2(I + J)$ on M_6 . Each of these decompositions is multiplicity-free, so the four carriers cover the five relevant irreducibles and an operator-norm bound on each carrier is exactly a bound on the irreducibles it contains. With $\hat{D}_3 = N/2^{64}$ for an integer matrix N , the script verifies in exact integer arithmetic that

$$71^2 2^{128} G - 2000^2 N^\top G N \succ 0$$

for every representative, using fraction-free Bareiss leading principal minors. Hence $\|\hat{D}_3\|_{\text{op}} \leq 71/2000 = 0.0355$ exactly. A second all-irrep pass certifies the tighter global upper bound 0.0354002321. For the bottleneck configuration, which lives on M_{41} , the script forms $S = N^\top G N$ and the pencil $p(y) = \det(yG - S)$; a Sturm chain isolates the largest root $y_{\max}$ of p in exact rational arithmetic, and $\sigma_{\max}^2 = y_{\max}/2^{128}$, which gives

$$0.035400232083798851 \dots \leq \max \|\hat{D}_3\|_{\text{op}} \leq 0.0354002321.$$

The full certificate format—carriers, Gram forms, orbit accounting, and the integer pencil—is documented with the artifact (CERTIFICATE_LEVEL3.md). Scripts 17 and 18 independently reproduce the same bottleneck numerically.

The contraction occurs because:

- If at least one sub-block has $\|\hat{D}_2\|_{\text{op}} < 1$, submultiplicativity gives $\|\hat{D}_3\|_{\text{op}} \leq \|\hat{D}_{2,A}\|_{\text{op}}^2 \cdot \|\hat{D}_{2,B}\|_{\text{op}}^2 < 1$.
- If both sub-blocks have $\|\hat{D}_2\|_{\text{op}} = 1$ but preserve different directions $v_A \neq v_B$ (guaranteed by Lemma 4), then $\hat{D}_{2,A}$ maps v_B to a vector with component $|\langle v_A, v_B \rangle| = 1/4$ along v_A , causing the product to contract. $\square$

4.6 Induction to All Levels

Theorem 9 (Exponential Contraction). *For every fixed valid balanced transposition-template program for point functions CC_T over $\{0, 1\}^\ell$ with $\ell = 2^d$, $d \geq 3$:*

$$\|\hat{D}_d((4, 1))\|_{\text{op}} \leq \lambda^{4^{d-3}},$$

where $\lambda = 0.0355$.

Proof. At depth $d + 1$, the commutator structure gives $\widehat{D}_{d+1} = \widehat{D}_A \cdot \widehat{D}_B \cdot \widehat{D}_A^\top \cdot \widehat{D}_B^\top$. The sub-blocks are valid whenever the root configuration is valid: if a sub-block target were e , its ancestor commutator would be $[e, \cdot] = e$, forcing the root target to be trivial. Thus the level- d bound applies to both children. By submultiplicativity of the operator norm and $\|M^\top\|_{\text{op}} = \|M\|_{\text{op}}$:

$$\begin{aligned} \|\widehat{D}_{d+1}\|_{\text{op}} &\leq \|\widehat{D}_A\|_{\text{op}} \cdot \|\widehat{D}_B\|_{\text{op}} \cdot \|\widehat{D}_A^\top\|_{\text{op}} \cdot \|\widehat{D}_B^\top\|_{\text{op}} \\ &= \|\widehat{D}_A\|_{\text{op}}^2 \|\widehat{D}_B\|_{\text{op}}^2 \\ &\leq (\lambda^{4^{d-3}})^4 = \lambda^{4^{(d+1)-3}}. \end{aligned}$$

The last inequality applies the uniform level- d induction hypothesis separately to the potentially different children A and B . The base case $\|\widehat{D}_3\|_{\text{op}} \leq \lambda$ is Theorem 8. Iterating: $\|\widehat{D}_d\|_{\text{op}} \leq \lambda^{4^{d-3}}$. $\square$

4.7 Why S_5 is Structurally Unique

Theorems 7 and 8 together provide a precise structural explanation of why S_5 is, among the symmetric groups we settle, the unique one where uniform level-3 spectral contraction succeeds in the standard representation. Three properties conspire:

1. **S_5 is non-solvable.** This makes it possible to choose non-commuting targets and conjugates so that the Barrington commutator construction does not collapse to the identity. It does not mean that arbitrary non-identity elements have nontrivial commutator.
2. **$\text{Stab}(k) \cong S_4$ is solvable with derived length exactly 3.** The derived series $S_4 \triangleright A_4 \triangleright V_4 \triangleright \{e\}$ terminates at depth 3, which is precisely the level of the commutator tree where contraction occurs. This forces the direction-mixing mechanism of Lemma 4.
3. **The preserved subspace dimension is at the critical half threshold.** $\widehat{D}_1$ preserves $n - 3 = 2$ of $n - 1 = 4$ dimensions, exactly half of the ambient space, allowing two preserved subspaces to be disjoint. For $n \geq 6$, the ratio $(n - 3)/(n - 1) > 1/2$ makes disjoint placement impossible (Theorem 7).

These properties are not independent coincidences. The first two express a single structural condition on the point stabilizer: a level-2 block preserves a direction v_k exactly when its transpositions fix k —that is, lie in $\text{Stab}(k) \cong S_{n-1}$ (Lemma 3)—and the direction-mixing mechanism that breaks v_k at level 3 is driven by the *solvability* of S_{n-1} : for S_5 , $\text{Stab}(k) \cong S_4$ is solvable and its derived length (3) is exactly the level at which contraction occurs. S_5 is the largest symmetric group whose point stabilizer is solvable, and the settled cases fit this picture— S_6, S_7 (non-solvable stabilizer) admit norm-1 configurations, and $n \geq 10$ is ruled out dimensionally. The present analysis does not decide $n \in \{8, 9\}$. This points past the symmetric groups: it suggests that commutator-tree mixing in a finite group may be governed not by the group’s size but by the solvability of the subgroups its leaves generate—a possible solvability barrier. Any extension of the present framework—to wider symmetric groups, or to any other finite group—must therefore confront this solvability structure rather than the size of the group, and must in each case specify the leaf class, the representation analysed, the fixed spaces of the resulting operators, and the subgroups the leaves generate.

5 Main Result

We now assemble the pieces—exact T -independence (Theorem 3), the conditioning lemma (Lemma 1), and the level-3 contraction with its inductive amplification (Theorems 8 and 9)—into the two quantitative guarantees.

Theorem 10 (Main Theorem). *For every fixed valid balanced transposition-template program for point functions CC_T over $\{0, 1\}^\ell$ with $\ell = 2^d$, $d \geq 3$:*

- (a) $D_T = D_{T'}$ for all $T, T' \in \{0, 1\}^\ell$.
- (b) $d_{\text{TV}}(D_T, \text{Uniform}(S_5)) \leq 6 \cdot \lambda^{4^{d-3}} + 2^{-\ell(\ell-1)}$, where $\lambda = 0.0355$ (the additive term is the conditioning loss of Lemma 1).
- (c) In particular, $d_{\text{TV}}(D_T, U) = \text{negl}(\ell)$.

Proof. Part (a) is Theorem 3. For part (b), the Fourier transform $\widehat{D}$ computed below is that of the unconditional distribution P_T , so by Theorem 2:

$$d_{\text{TV}}(P_T, U)^2 \leq \frac{1}{4} \sum_{\rho \neq \text{triv}} d_\rho \cdot \|\widehat{D}(\rho)\|_{\text{F}}^2.$$

By Remark 5, the sign representation contributes zero. For each remaining representation ρ of dimension d_ρ , Proposition 1 gives a base case $\|\widehat{D}_3(\rho)\|_{\text{op}} \leq \lambda_\rho < 1$ for all valid level-3 configurations. The submultiplicativity argument of Theorem 9 applies to each ρ independently (it uses only $\|M^\top\|_{\text{op}} = \|M\|_{\text{op}}$ and submultiplicativity, which hold in any representation), yielding $\|\widehat{D}_d(\rho)\|_{\text{op}} \leq \lambda_\rho^{4^{d-3}}$. Since $\|M\|_{\text{F}}^2 \leq d_\rho \cdot \|M\|_{\text{op}}^2$:

$$\sum_{\rho} d_\rho \cdot \|\widehat{D}(\rho)\|_{\text{F}}^2 \leq \sum_{\rho} d_\rho^2 \cdot \lambda_\rho^{2 \cdot 4^{d-3}} \leq \left(\sum_{\rho} d_\rho^2 \right) \cdot \lambda^{2 \cdot 4^{d-3}},$$

where the exact global certificate gives $\max_{\rho} \lambda_{\rho} \leq 0.0354002321 < 0.0355$ and the standard representation contains the certified lower endpoint $0.035400232083798851 \dots$ (Proposition 1 and Theorem 8). We use the conservative $\lambda = 0.0355$. Since $\sum_{\rho} d_\rho^2 = 4^2 + 5^2 + 6^2 + 5^2 + 4^2 = 118$:

$$d_{\text{TV}}(P_T, U)^2 \leq \frac{118}{4} \cdot \lambda^{2 \cdot 4^{d-3}}, \quad d_{\text{TV}}(P_T, U) \leq \sqrt{\frac{118}{4}} \cdot \lambda^{4^{d-3}} < 6 \cdot \lambda^{4^{d-3}}.$$

Finally, by Lemma 1,

$$d_{\text{TV}}(D_T, U) \leq d_{\text{TV}}(P_T, U) + 2^{-\ell(\ell-1)} < 6 \cdot \lambda^{4^{d-3}} + 2^{-\ell(\ell-1)}.$$

For part (c), since $\lambda < 1$ and 4^{d-3} grows exponentially in $d = \log_2 \ell$ (so $\lambda^{4^{d-3}}$ decays doubly-exponentially in d , i.e. like $\exp(-\Theta(\ell^2))$ in ℓ), the bound is negligible in ℓ . $\square$

5.1 The fresh-sample terminal-output experiment

The Main Theorem controls one exactly specified experiment. We record its q -fold form, since that is the form a reduction would consume, and then delimit precisely what it does and does not cover.

Definition 7 (Fresh-sample terminal-output transcript). Fix $\ell = 2^d$ with $d \geq 3$ and a target $T \in \{0, 1\}^\ell$. The q -sample terminal-output transcript of BP_T is $(\pi_1, \dots, \pi_q)$, where each $\pi_i \in S_5$ is the terminal output permutation of BP_T evaluated on a *fresh, independent, uniformly random inconsistent path*; the π_i are thus i.i.d. with law D_T , and we write $D_T^{\otimes q}$ for the law of the transcript.

Corollary 1 (Terminal-output indistinguishability). Let $\varepsilon(\ell) = 6\lambda^{4^{d-3}} + 2^{-\ell(\ell-1)}$ be the bound of Theorem 10(b), with $\lambda = 0.0355$. For every integer $q \in \mathbb{Z}_{\geq 1}$ and all $T, T' \in \{0, 1\}^\ell$:

- (a) $D_T^{\otimes q} = D_{T'}^{\otimes q}$ (exactly, not merely statistically);
- (b) $d_{\text{TV}}(D_T^{\otimes q}, \text{Uniform}(S_5)^{\otimes q}) \leq \min\{1, q\varepsilon(\ell)\}$;
- (c) for every target-independent randomised post-processing K — a Markov kernel from S_5^q to an arbitrary finite space, fixed without knowledge of T — both conclusions persist: $KD_T^{\otimes q} = KD_{T'}^{\otimes q}$ and $d_{\text{TV}}(KD_T^{\otimes q}, K\text{Uniform}(S_5)^{\otimes q}) \leq \min\{1, q\varepsilon(\ell)\}$.

Proof. (a) By Theorem 10(a) the coordinate laws coincide, $D_T = D_{T'}$; the coordinates are independent, hence the product laws coincide.

(b) Let H_j be the hybrid law whose first j coordinates are uniform on S_5 and whose remaining $q - j$ coordinates are i.i.d. D_T , so that $H_0 = D_T^{\otimes q}$ and $H_q = \text{Uniform}(S_5)^{\otimes q}$. Consecutive hybrids differ in one coordinate only, and total variation between product measures differing in a single independent coordinate equals the total variation of that coordinate, so $d_{\text{TV}}(H_j, H_{j+1}) = d_{\text{TV}}(D_T, \text{Uniform}(S_5)) \leq \varepsilon(\ell)$ by Theorem 10(b). Telescoping with the triangle inequality over $j = 0, \dots, q - 1$ gives $d_{\text{TV}}(D_T^{\otimes q}, \text{Uniform}(S_5)^{\otimes q}) \leq q \varepsilon(\ell)$; the bound 1 is trivial.

(c) Total variation contracts under Markov kernels (data processing): $d_{\text{TV}}(K\mu, K\nu) \leq d_{\text{TV}}(\mu, \nu)$ for every kernel K and all laws μ, ν . Since K is fixed independently of T , part (a) gives $KD_T^{\otimes q} = KD_{T'}^{\otimes q}$, and data processing applied to part (b) gives the stated bound. $\square$

Remark 7 (Scope of Corollary 1). Corollary 1 concerns the terminal-output marginal under *fresh, independent, uniformly random* inconsistent paths, together with target-independent post-processing of that transcript. Its hypotheses distinguish this experiment from three other kinds of observation:

- fixed, adversarially chosen, or adaptive paths, including choices based on earlier outputs;
- correlated evaluations, such as evaluations sharing path bits;
- intermediate products or an encoded adversarial view, including Kilian randomisation, encoded matrices, and graded encodings.

Applying the result to any of these settings requires an additional argument relating that observation to the stated experiment. The construction-level example in Section 1 describes this requirement for obfuscation.

Remark 8 (Non-power-of-2 input lengths). The theorem is stated only for the complete balanced tree with $\ell = 2^d$. Padding by leaves that output the identity in both branches is not innocuous in a commutator tree: a subcommutator of the form $[g, e]$ collapses to the identity and can change the target of the parent subtree. Extending the result to arbitrary ℓ therefore requires a separate template and a separate count of consistent and inconsistent paths.

For concrete values:

ℓ	Depth d	$\ \widehat{D}_{(4,1)}\ _{\text{op}}$ bound	d_{TV} bound
8	3	3.55×10^{-2}	2.13×10^{-1}
16	4	1.59×10^{-6}	9.53×10^{-6}
32	5	6.36×10^{-24}	3.82×10^{-23}
64	6	1.64×10^{-93}	9.83×10^{-93}

6 Extensions and Open Problems

6.1 Conjunctions with Wildcards

A conjunction $\text{CC}_{T,W}$ with target $T \in \{0,1\}^\ell$ and wildcard set $W \subseteq [\ell]$ outputs 1 iff $x_i = T_i$ for all $i \notin W$. In the branching program, wildcard positions output σ_i for both bit values. Obfuscation of conjunctions has been studied extensively [BVWW16, BKM⁺18, BLMZ19], which is what motivates looking at this class here. For conjunctions we prove T-independence of the terminal marginal and report computational evidence for mixing.

Theorem 3 extends directly to conjunctions: $D_{T,W} = D_{T',W}$ for all T, T' with the same wildcard set W . The spectral contraction depends on the wildcard density $|W|/\ell$. Computational evidence shows strong contraction for $|W|/\ell \leq 1/2$, weakening contraction for $|W|/\ell = 3/4$, and no contraction for $|W|/\ell = 1$ (the trivial function). The precise threshold is an open problem.

6.2 Beyond Point Functions

For general branching programs computing the same Boolean function via different constructions, “strong universal mixing” ($D_{C_0} = D_{C_1}$) is **false**: programs with different transposition assignments produce different distributions. However, “weak universal mixing” (both distributions converge to uniform independently) appears to hold: computational evidence shows equivalent circuits with different pair assignments achieving similar d_{TV} values. Proving this would extend the algebraic terminal-output property to a larger class of programs.

Computational experiments reveal a stronger pattern. For $\ell = 4$, two programs computing AND_4 with different transposition pairs yield Fourier norms $\|\widehat{D}(4, 1)\|_{\mathbb{F}}^2 = 0.147$ and 0.293 respectively — clearly different distributions — yet *identical* total variation distances $d_{TV} = 0.309$ to uniform. A sampled OR-of-AND experiment shows approximately equal distances for two structurally different implementations ($d_{TV} = 0.1904$ versus 0.1923). If this “equidistance” phenomenon is general, it would mean that $d_{TV}(D_C, U)$ depends only on the Boolean function computed, not on the specific Barrington construction — a property strictly stronger than weak universal mixing.

6.3 Generalization to Other Groups

The dimensional obstruction in Theorem 7 prevents a direct extension of the S_5 direction-mixing proof to S_n with $n \geq 6$. This motivates studying other representations, deeper levels of the commutator tree, and other groups. Finite matrix groups such as $\text{GL}_n(\mathbb{F}_q)$ are natural mathematical candidates for such a move. The obstruction of Theorem 7 is specific to the transposition leaf class and to the standard representation of S_n ; it therefore does not decide the matrix case, which would first require fixing a leaf class and a representation before the analogous fixed-space count could be stated.

Our proof follows a three-step program that is, in principle, portable: (1) define the output distribution on inconsistent inputs, (2) identify the bottleneck irreducible representation, (3) discover geometric structure in the spectral obstructions. For $\text{GL}_2(\mathbb{F}_q)$ with q a small prime, the irreducible representations are completely classified (principal series, cuspidal, Steinberg, and one-dimensional characters), and the analogue of step (2) is computationally feasible. Determining whether the level-2 obstructions for matrix branching programs over $\text{GL}_2(\mathbb{F}_q)$ exhibit geometric regularity analogous to the simplex in Observation 6 would be a concrete first test of whether spectral contraction extends beyond symmetric groups.

Establishing contraction in every nontrivial irreducible would yield an unconditional mixing theorem for such a template over a matrix group. A cryptographic application would then require the construction-specific connection described in Section 1.

6.4 Deeper Levels and the Solvability Barrier

Two independent obstructions make S_5 special in the standard representation, and extending beyond S_5 requires confronting both.

The first, discussed in Section 4.7, is dimensional (Theorem 7): for $n \geq 6$, the preserved subspace of $\widehat{D}_1$ has dimension $n - 3$ in $\mathbb{R}^{n-1}$, forcing every level-2 quadruple to have $\|\widehat{D}_2\|_{\text{op}} = 1$, which blocks level-3 operator-norm contraction from following the same direction-mixing proof used for S_5 .

The second is algebraic. The proof of Lemma 4 relies on the solvability of S_4 : the derived series $S_4 \triangleright A_4 \triangleright V_4 \triangleright \{e\}$ forces level-2 targets into V_4 (abelian), killing the outer commutator. For S_n with $n \geq 6$, the relevant stabilizer is S_{n-1} , which is *non-solvable*, so even if the dimensional obstruction could be sidestepped, the direction-mixing argument would require a fundamentally different mechanism.

The analysis presented here leaves the following questions undecided. For S_6 , S_7 , and $n \geq 10$, where uniform contraction in the standard representation fails at level 3, one may ask whether a uniform bound $\|\widehat{D}_d\|_{\text{op}} < 1$ holds at some level $d \geq 4$. For S_8 and S_9 , the corresponding question starts at level 3. A separate path, for every $n \geq 6$, is to study *non-standard representations*: whether some irreducible representation other than $(n-1, 1)$ contracts at level 3. Note that a Diaconis–Shahshahani mixing bound requires controlling *all* nontrivial irreducible representations simultaneously, so contraction in a single non-standard representation alone does not yield mixing; the bottleneck may differ from $(n-1, 1)$. These constitute the most natural follow-up directions.

7 Computational Verification

The proofs of Theorems 4 and 5 use exact symbolic algebraic arithmetic (implemented in `sympy`) with a verified orthonormal basis for the standard representation: that basis carries the radicals $\sqrt{2}$, $\sqrt{6}$, $\sqrt{12}$, $\sqrt{20}$, which `sympy` represents exactly, so the intermediate quantities are exact algebraic numbers (the reported Frobenius norms are rational). Theorem 8 uses an exact integer Gram-form certificate over the full ordered universe of level-3 configurations—9,447,840 oriented pairs across all five non-trivial, non-sign representations, partitioned into 39,591 symmetry orbits with exact accounting (Script 20). Scripts 17 and 18 provide independent full-universe `numpy` cross-checks.

The reproducibility archive is available at <https://github.com/ernest-zermeno/uniform-mixing-barrington-branching-programs>. It contains all 20 verification scripts, the exact level-3 certificate, analytic proof notes, dependencies, and instructions for reproducing the checks.

Script	Content	Method
01	Ambivalence of S_5 (similarity blindness)	Exhaustive
02	Collision analysis, $\ell = 4$	Exhaustive
03	Collision scaling, $\ell = 4 \rightarrow 8$	Sampled
04	Fourier spectral analysis	Exact + sampled
05	Barrington construction + Kilian obfuscation	Exhaustive
06	Contraction operator (unconditional distribution)	Analytic
07	T-independence proof verification	Exhaustive
08	Conjunctions with wildcards	Exhaustive
09	Universal mixing investigation	Exact + sampled
10	Complete spectral contraction proof	Exact + numerical
11	Eigenvector analysis (simplex discovery)	Numerical
12	Reduced level-3 census, standard representation	Numerical
13	Reduced level-3 census, all representations	Numerical
14	Obstruction checks (S_5, S_6, S_7)	Numerical + exact
15	Exact sympy proof of singular values of $\widehat{D}_1$	Exact
16	Conditioning lemma, $\ell = 4$	Exact
17	Ordered-orientation level-3 cross-check	Numerical
18	All-irreps ordered-orientation cross-check	Numerical
19	Ordered level-2 exact cross-check	Exact
20	Full-universe level-3 Gram certificate	Exact integer

Scripts 10–13 provide reduced-orientation cross-checks for S_5 ; their numerical maxima are not uniform bounds over the full ordered universe. The level-3 census in Scripts 12–13 contains 73,810 retained pairs. Scripts 17 and 18 cross-check all 9,447,840 orientations numerically. Script 20 supplies the definitive proof-grade certificate: exact symmetry accounting followed by integer Gram-form inequalities for every relevant irreducible proves

the $71/2000$ bound and the tighter global upper bound 0.0354002321 . Scripts 14 and 15 verify Theorem 7: Script 14 exhaustively enumerates the reduced level-2 and level-3 censuses of S_6 and S_7 , reporting a maximum norm of 1 and also strictly contractive configurations. It verifies explicit norm-1 examples and exact Gram-form certificates for contractive examples in both groups. Script 15 gives the exact singular-value pattern $[1, \dots, 1, 1/8, 0]$ of $\widehat{D}_1$ for S_5, S_6, S_7 . Script 16 verifies the conditioning lemma for $\ell = 4$ over all 2^{16} paths in exact rational arithmetic. Scripts 01–09, 11–13, 17–19 provide additional cross-validation.

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